Ilya Haider

Email: ilyahaider15313@gmail.com
LinkedIn: www.linkedin.com/in/ilya-haider-aa35a2385
GitHub: https://github.com/settings/profile
Kaggle: https://www.kaggle.com/sanachips
Personal Website: https://portfolio-website-vlso.onrender.com
Mailing Address: KIU, Gilgit, Pakistan
Phone Number: +923494788714

Objective & Research Interest

Highly motivated to engage in cutting-edge research within a collaborative lab environment. Aiming to strengthen technical and analytical skills through hands-on investigation of real-world problems. Curious and committed to continuous learning, with the goal of growing as an independent and impactful researcher.

Key Research Interests: Computer Vision, Deep Learning for Medical & Natural Scenes, NLP, Machine Learning.

Education

Bachelors in Software Engineering **Aug 2020 – Jun 2024**
Government College University Faisalabad, Pakistan
CGPA: 3.32/4.0
Recognized as a top ten performer for 2020-2024 at GCUF
BS Thesis Project: "*Comparative analysis of BERT and XLNet for text classification*"
Major Subjects: Artificial Intelligence, Natural Language Processing, Data structure, Design & Analysis of Algorithm, Database System, Statistical theory, Calculus & Analytics, Discrete structure, Human computer Interaction

Bachelor of Education (B.Ed. 1.5 Year Program) **Apr2026 – Present**
Virtual University of Pakistan
Remotely enrolled in teacher education and professional development

Intermediate ICS (Computer Science) **Mar 2018 – Jun 2020**
Kauthar College for Women Islamabad, Pakistan
Grade: A+ (898/1100)
Major Subjects: Computer Sciences, Mathematics, Physics, English, Urdu, Islamiyat & Pakistan Studies

Publications

Title: *"Artificial Intelligence for Brain Tumor Classification Using MRI Imaging"*
Under Review in HEC-recognized Y Category journal.
The research explores the use of Artificial Intelligence, Machine Learning, and Deep Learning models for automated brain tumor detection and classification using MRI scans. Worked on studying CNN-based approaches, transfer learning techniques, and medical image preprocessing to improve classification accuracy and diagnostic support systems. The study contributes to the fields of Medical Imaging, Radiology, Biomedical Engineering, and Computer Science.

Work Experience

Visiting Faculty (Computer Science) at Karakoram International University (KIU) Gilgit, Pakistan **Mar 2025 – Present**
Instructing Course: Machine Learning, Software Engineering, Information Security, Database, Data Privacy & Encryption Database System and Computer Organization & Assembly Language with supporting labs.

Supervising Paid/Semester Projects: Clinic Management System, Shop Management System, Blood Bank Management System, Airline Reservation Management System, School Management System, Library Management System.

Computer Science Instructor at Gilgit Army Public School & College (GAPS&C), Pakistan **Oct 2025 – Present**

Instructing Computer Sciences to the students of Intermediate

Remote Research Contributor at Sejong University, South Korea **Aug 2025 – Present**

Strengthening understanding of deep learning models for classification detection & segmentation, model evaluation, and research methodologies.

Image Segmentation Intern at Machine Vision & Intelligent Systems Lab, Islamabad, Pakistan **Feb 2023 – Apr 2023**

Model (CNN) training using pre-trained models. Evaluated models using IoU, Dice, Pixel Accuracy, and presented results through visual reports

Cyber Security Intern at IT department of Sadaqat Pvt Limited, Faisalabad, Pakistan **Mar 2022 – Sep 2022**

Collaborated with IT team to install different software, monitored networks and running protocols

Research Projects

Comparison of BERT & XLNet for Text Classification (Final Year Project)

- Conducted a comparative study of transformer-based NLP models, BERT and XLNet, for text classification tasks.
- Fine-tuned models on multiple benchmark datasets for sentiment analysis and spam detection.
- Evaluated model performance and achieved up to 91% accuracy with BERT and 92% with XLNet, highlighting improved contextual representation in XLNet for complex sequences.

Performance Evaluation of Pretrained CNN Models on Tomato Image Dataset

- Applied transfer learning techniques using pretrained CNN architectures including VGG, ResNet, GoogLeNet, and U-Net.
- Performed image preprocessing, augmentation, and dataset splitting for robust model training.
- Evaluated models using accuracy metrics, confusion matrices, and loss analysis.
- Achieved highest accuracy of 94% using ResNet, outperforming VGG and GoogLeNet by 6–8% on validation data.

Human-Centered Sentiment Analysis Using Deep Learning

- Built a sentiment analysis system using deep learning models including LSTM and BERT.
- Applied NLP preprocessing techniques such as tokenization, text normalization, and feature extraction.
- Evaluated model performance using precision, recall, F1-score, and accuracy metrics.
- Achieved 88% accuracy with LSTM and improved performance up to 93% using BERT fine-tuning.

Enhanced Moving Object Detection in Static Video Streams Using Multiple Median Model

- Designed and implemented a moving object detection system for static video streams using the Multiple Median Model.
- Compared system performance with Gaussian Mixture Model (GMM) and Multiple Mean Model approaches.
- Evaluated detection accuracy on Waving Trees, Bootstrapping, and Outdoor datasets.
- Improved detection accuracy to 87% with reduced background noise sensitivity compared to GMM baseline.

Machine Learning-Based Customer Churn Prediction in Telecom Sector

- Developed a predictive analytics system to identify high-risk telecom customers likely to churn.
- Performed preprocessing tasks including missing value handling, feature encoding, and class balancing using Python libraries.

- Implemented Logistic Regression, Random Forest, and XGBoost models using Scikit-learn.
- Achieved strong predictive performance with 89% accuracy and 0.92 AUC score.
- Visualized feature importance and model evaluation results for business decision support.

Real-Time Static Sign Language Recognition Using CNN

- Developed a CNN-based sign language recognition system using the Sign Language MNIST dataset.
- Applied image preprocessing and augmentation techniques to improve model generalization.
- Trained and evaluated the model for accurate classification of static hand gestures representing alphabet signs.
- Achieved 95% classification accuracy with improved generalization on unseen gesture samples**.**

Conferences & Seminars

Two-Day Workshop on Current Topics in Machine Learning and Data Analysis
Organized by: School of Science and Engineering, LUMS.
Topics Covered: Machine Learning for Materials Design, Topological Data Analysis and Financial Time Series Analysis.

2-Day Hands-on Training Workshop on Data Analysis & Machine Learning
Organized by: COMSATS University Islamabad.
Focus: Practical training in Data Analysis and Machine Learning techniques

Seminar on "How to Submit a Research Paper"
Organized by: The University of Faisalabad (TUF).
Focus: Research methodology and best practices in academic publishing.

Skills

Programming	Python, C++, R, Markdown, HTML, CSS, JavaScript, PHP
Libraries	PyTorch, TensorFlow, Keras, OpenCV, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, SciPy
Software & Tools	Anaconda, PyCharm, VS Code, RStudio, Google Colab, NetBeans, Kali Linux, Github, Kaggle, Google Colab, Jupyter Notebook

Certifications

Machine Learning with Python – **Coursera**
Databases and SQL with Python – **Coursera**
Certified in Cyber Security - **NAVTTC** PM's Youth Skill Development
Introduction to IOT - **LearnOBots** Islamabad

Awards

KCW Meritorious Scholar: Acknowledged for outstanding performance by KCW.
IEEE Winner: Honored for excellence in programming quiz,
Secured 1st position: In book reading comprehension and grammar.
Silver Medalist: Attained second place at the prestigious Dolphin Camps season two & three.

References

Dr. Muhammad Awais (BS Thesis Supervisor)
Chairperson (Head of Department)
Department of Software Engineering
Government College University, Faisalabad, Pakistan
Email: muhmmadawais@gcuf.edu.pk
Cell #: +92 336 1161635

Dr. Muhammad Yahya Saeed
Assistant Professor
Department of Software Engineering
Government College University, Faisalabad, Pakistan
Email: m_yahya_saeed@gcuf.edu.pk
Cell #: +92 300 6649235